Unit: 6 Unit 4: Absolute Value **Name:** ________________

Topic: Absolute Value Inequalities and Their Applications **Date:** __________

Challenge (Habanero)

Q1. Solve $|x-2|>5$.
(A) $x<-3$ or $x>7$
(B) $x<-3$ and $x>7$
(C) $x>-3$ or $x<7$
(D) $x>-3$ and $x<7$
(E) $x=-3$ or $x=7$

Q2. A hiker wants to be within 4 miles of the campsite. If x represents the distance from the hiker to the campsite, the inequality that represents this situation is
(A) $|x|leq4$
(B) $|x|geq4$
(C) $|x|>4$
(D) $|x|<4$
(E) $x=4$

Q3. The population of a certain animal species is given by $|x-10|leq5$, where x is the number of years since the species was first introduced. What is the range of years when the population is within 5 units of 10?
(A) $5leqxleq15$
(B) $5<x<15$
(C) $0leqxleq20$
(D) $0<x<20$
(E) $10leqxleq20$

Q4. A car's fuel consumption is given by $|x-25|geq5$, where x is the speed in miles per hour. What speeds will result in fuel consumption outside the range of 5 units from 25?
(A) $x<20$ or $x>30$
(B) $x<20$ and $x>30$
(C) $x>20$ or $x<30$
(D) $x>20$ and $x<30$
(E) $x=20$ or $x=30$

Q5. The graph of $|x+3|<2$ is
(A) a single point at $x=-1$
(B) an open interval between $x=-1$ and $x=-5$
(C) a closed interval between $x=-1$ and $x=-5$
(D) an open interval between $x=-5$ and $x=-1$
(E) the entire number line

Q6. Solve $|2x-5|leq3$.
(A) $1leqxleq4$
(B) $1<x<4$
(C)
rac12 leq x leq
rac72
(C)
rac12 < x <
rac72
(C) $x=1$ or $x=4$

Q7. The absolute value inequality $|x-4|>2$ can be written in interval notation as
(A) $(-infty,2)cup(6,infty)$
(B) $(-infty,2]cup[6,infty)$
(C) $(2,6)$
(D) $[2,6]$
(E) $(-infty,infty)$

Q8. A park ranger wants to ensure that the temperature is within 10^circF of 50^circF. If x represents the temperature, the inequality that represents this situation is
(A) $|x-50|leq10$
(B) $|x-50|geq10$
(C) $|x-50|>10$
(D) $|x-50|<10$
(E) $x=50$

Open Ended Questions (FRQ)

Q9. A certain species of bird migrates to areas where the temperature is between 40^circF and 80^circF. If x represents the temperature in degrees Fahrenheit, write an absolute value inequality that represents the temperatures that the birds will migrate to, and graph the solution set on a number line.

Q10. A car's fuel efficiency is given by $|x-30|leq5$, where x is the speed in miles per hour. What speeds will result in fuel efficiency within 5 units of 30 miles per gallon? Write your answer in interval notation and graph the solution set on a number line.

Chef's Tips

- Remind students to consider both the positive and negative cases when solving absolute value inequalities.
- Emphasize the importance of graphing the solution set on a number line.
- Encourage students to use real-world examples to illustrate the applications of absolute value inequalities.